CAE 2 – IoT Practical Examination

Programme: MSc / MCA Semester 3
Course: Internet of Things
Time: 3 Hours
Maximum Marks: 60

Instructions

- Answer **either Q1 or Q2** from Section A.
- Answer **either Q3 or Q4** from Section B.
- Demonstrate the working of the developed application with appropriate output.
- Students may use suitable Python libraries and tools as required.

SECTION A

Answer either Q1 or Q2

Q1. DHT22 Temperature and Humidity Monitoring

[30 Marks]

Develop an **ESP32-based temperature and humidity monitoring system** using a **DHT22 sensor**.

The system should:

1. Interface the DHT22 sensor with ESP32.
2. Read the **temperature and humidity** values.
3. Display the sensor readings through the **Serial Monitor**.
4. Demonstrate the working of the system with suitable sensor readings.

OR

Q2. LDR-Based Light Intensity Monitoring

[30 Marks]

Develop an **ESP32-based light intensity monitoring system** using an **LDR sensor**.

The system should:

1. Interface the LDR sensor with ESP32.
2. Read the **light intensity** from the sensor.
3. Display the sensor readings through the **Serial Monitor**.
4. Demonstrate the working of the system with suitable sensor readings.

SECTION B

Answer either Q3 or Q4

Q3. CoAP-Based Air Quality Monitoring and Random Forest Classification

[30 Marks]

Develop a CoAP-based air quality monitoring and classification system for Kochi.

Input API Link:
https://air-quality-api.open-meteo.com/v1/air-quality?latitude=9.93&longitude=76.27&current=pm10,pm2_5,nitrogen_dioxide,sulphur_dioxide,ozone

The system should use the following air-quality parameters:

PM2.5, PM10, NO_2, SO_2 and O_3

Dataset:
The Random Forest model should be trained using historical air-quality data from the following Kaggle dataset:

Official Kaggle Dataset:
https://www.kaggle.com/datasets/rohanrao/air-quality-data-in-india

File: `city_day.csv`

The system should:

1. Develop a CoAP Server and CoAP Client. [6 Marks]

2. On receiving a request from the CoAP Client, retrieve the live air-quality data of Kochi from the given Air Quality API through the CoAP Server. [6 Marks]
3. Transfer the retrieved air-quality data from the CoAP Server to the CoAP Client using a CoAP GET request. [5 Marks]
4. Develop a dashboard to display the retrieved air-quality parameters (PM2.5, PM10, NO_2, SO_2 and O_3). [5 Marks]
5. Train and apply a Random Forest classification model using the historical `city_day.csv` dataset to classify the current air-quality data into appropriate air-quality categories such as Good, Satisfactory, Moderate, Poor, Very Poor, or Severe. [5 Marks]
6. Display the predicted air-quality classification on the dashboard. [3 Marks]

The system should demonstrate the following flow:

Air Quality API → CoAP Server → CoAP Client → Dashboard → Random Forest Classification

The Random Forest model should use:

Input Features: PM2.5, PM10, NO_2, SO_2 and O_3
Target: `AQI_Bucket`

Live air quality view : https://app.cpcbccr.com/

OR

Q4. CoAP-Based Weather Monitoring and Decision Tree Classification

[30 Marks]

Develop a CoAP-based weather monitoring and classification system for Kochi.

Input API Link:

https://api.open-meteo.com/v1/forecast?latitude=9.93&longitude=76.27&current=temperature_2m,relative_humidity_2m,precipitation,cloud_cover,wind_speed_10m,surface_pressure

The system should retrieve the following weather parameters:

- Temperature
- Relative Humidity
- Precipitation
- Cloud Cover
- Wind Speed
- Surface Pressure

Dataset:

The Decision Tree classification model should be trained using historical weather data retrieved from the Open-Meteo Historical Weather API.

Historical Dataset API:

https://archive-api.open-meteo.com/v1/archive?latitude=9.93&longitude=76.27&start_date=2023-01-01&end_date=2025-12-31&hourly=temperature_2m,relative_humidity_2m,precipitation,cloud_cover,wind_speed_10m,surface_pressure,weather_code

The historical dataset should contain the following parameters:

- `temperature`
- `humidity`
- `precipitation`
- `cloud_cover`
- `wind_speed`
- `surface_pressure`
- `weather_code`

The `weather_code` should be converted into suitable weather classes such as:

- Clear
- Cloudy
- Drizzle
- Rain

The system should:

1. Develop a CoAP Server and CoAP Client. [6 Marks]
2. On receiving a request from the CoAP Client, retrieve the live weather data of Kochi from the given Open-Meteo API through the CoAP Server. [6 Marks]
3. Transfer the retrieved weather data from the CoAP Server to the CoAP Client using a CoAP GET request. [5 Marks]

4. Develop a dashboard to display the retrieved weather parameters: Temperature, Relative Humidity, Precipitation, Cloud Cover, Wind Speed and Surface Pressure. [5 Marks]
5. Train and apply a Decision Tree classification model using the historical weather dataset to classify the current weather conditions into appropriate weather categories such as Clear, Cloudy, Drizzle or Rain. [5 Marks]
6. Display the predicted weather classification on the dashboard. [3 Marks]

The system should demonstrate the following flow:

Open-Meteo Weather API → CoAP Server → CoAP Client → Dashboard → Decision Tree Classification

The Decision Tree model should use:

Input Features:

- Temperature
- Relative Humidity
- Precipitation
- Cloud Cover
- Wind Speed
- Surface Pressure

Target:
`weather_class`

The `weather_class` should be derived from the historical `weather_code` values.

weather_code	weather_class
0, 1	**Clear**
2, 3	**Cloudy**
51, 53, 55	**Drizzle**

61, 63, 65 **Rain**